Kirill Bykov, Ph.D.

Machine Learning Post-Doctoral Researcher | Mechanistic Interpretability | Vision-Language Models

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PROFILE

Postdoctoral researcher in Prof. Akata's group at TUM, developing methods for Mechanistic Interpretability for LLMs. Ph.D. with summa cum laude from TU Berlin; research spans neuron- and representation-level analysis, multimodal / language-facing interpretability, model auditing, and reliable evaluation of explanations.

WORK EXPERIENCE

Postdoctoral Researcher Aug 2025–Present
Technical University of Munich · Munich, Germany
Research in mechanistic interpretability for modern deep networks, with a focus on representations, feature reliability, and explanation methods for vision and language models.

Ph.D. Researcher in Machine Learning Jan 2021–Jun 2025
TU Berlin and ATB Potsdam · Berlin and Potsdam, Germany
Research in mechanistic interpretability and explainable AI, developing methods to analyze neural representations, identify learned concepts, detect spurious correlations, and audit anomalous model behavior.

Student Research Assistant May 2020–Jan 2021
Machine Learning Group, TU Berlin · Berlin, Germany
Research in explainable AI and Bayesian neural networks, contributing to methods for model explanations and uncertainty-aware analysis of deep networks.

Data Science Research Intern Dec 2019–Mar 2020
TomTom, AI Geospatial Research · Berlin, Germany
Research and analytics in geospatial machine learning, developing anomaly detection models for large-scale spatial data.

Data Scientist Jun 2018–Apr 2019
Skyeng · Remote
Research and analytics in recruitment and operations, building ML-driven candidate scoring, workload forecasting, process mining, and workflow optimization tools.

Data Analyst Sep 2015–Dec 2017
MegaFon · Saint Petersburg, Russia
Analytics in trade marketing and business operations, focusing on sales forecasting, operational reporting, performance monitoring, and process optimization.

EDUCATION

Ph.D. in Machine Learning 2021–2025
Technische Universität Berlin · Berlin, Germany
Advisors: Prof. Marina Höhne and Prof. Klaus-Robert Müller; thesis: *Explaining Representations in Deep Neural Networks*; grade: summa cum laude, 1.0.

M.Sc. in Data Science / ICT Innovation 2018–2020
TU Berlin and TU Eindhoven · Berlin, Germany / Eindhoven, the Netherlands
Double-degree program between TU Berlin, Germany, and TU Eindhoven, the Netherlands; EIT Digital Excellence Scholarship; grade: cum laude, 1.3.

B.Sc. in Applied Mathematics and Computer Science 2014–2018
Saint Petersburg State University · Saint Petersburg, Russia
Faculty of Mathematics and Mechanics, Department of Statistical Modelling; grade: 4.47 / 5.

SELECTED PUBLICATIONS

NeurIPS 2025	**Capturing Polysemanticity with PRISM: A Multi-Concept Feature Description Framework** L. Kopf, N. Feldhus, **K. Bykov**, P. L. Bommer, A. Hedström, M. M.-C. Höhne, O. Eberle
NeurIPS 2025	**Manipulating Feature Visualizations with Gradient Slingshots** D. Bareeva, M. M.-C. Höhne, A. Warnecke, L. Pirch, K.-R. Müller, K. Rieck, S. Lapuschkin, **K. Bykov**
NeurIPS 2024	**CoSy: Evaluating Textual Explanations of Neurons** L. Kopf, P. L. Bommer, A. Hedström, S. Lapuschkin, M. M.-C. Höhne, **K. Bykov**
NeurIPS 2023	**Labeling Neural Representations with Inverse Recognition** **K. Bykov**, L. Kopf, S. Nakajima, M. Kloft, M. M.-C. Höhne
TMLR 2023	**DORA: Exploring Outlier Representations in Deep Neural Networks** **K. Bykov**, M. Deb, D. Grinwald, K.-R. Müller, M. M.-C. Höhne
AAAI 2022	**NoiseGrad: Enhancing Explanations by Introducing Stochasticity to Model Weights** **K. Bykov**, A. Hedström, S. Nakajima, M. M.-C. Höhne
TMLR 2025	**Explaining Bayesian Neural Networks** **K. Bykov**, M. M.-C. Höhne, A. Creosteanu, K.-R. Müller, F. Klauschen, S. Nakajima, M. Kloft

LEADERSHIP, MENTORING, AND SERVICE

- **Research supervision:** Supervised 4 students; resulting work includes NeurIPS 2024, NeurIPS 2025, and an award-winning thesis recognized with the 2024 Rolf Niedermeier Prize.
- **Research organization:** Organized the "Global and Concept-Based Explainability" track at XAI 2024; moderated sessions at IEEE SaTML 2026, ECAI 2023.
- **Peer review and service:** Program Committee member for IEEE SaTML 2025, 2026; reviewer for NeurIPS, TMLR, IEEE TNNLS, and IEEE TRAMPI.
- **Grant writing:** Contributed to a successful BMBF-funded grant proposal (REFRAME, 01IS24073A-C) .

SELECTED INVITED TALKS

GLOBE: Global Explanation of Deep Neural Networks	Marburg University	Jun 2025
Explainable AI: From Local to Global	Potsdam Graduate School	Oct 2024
Labeling Neural Representations with Inverse Recognition	BLISS Berlin	Jan 2024
Panel discussion on Fair and Trustworthy AI	Helmholtz AI Conference	Jun 2022

TECHNICAL STRENGTHS

Technical	Python, PyTorch, vLLM, Hugging Face, SQL / NoSQL, Linux, Git, Bash, GCP / Azure, LLM coding agents, Codex
Languages	English C2, German B1, Russian native

SELECTED DISTINCTIONS

- Ph.D. defended with **summa cum laude**; thesis: *Explaining Representations in Deep Neural Networks*.
- EIT Digital Excellence Scholarship recipient, 2018–2020.
- Winner / prize-winner in hackathon, and mathematics competitions, including Data Natives 2019, HackDelft 2019, BioHack 2018, the SkolTech Statistical Learning Olympiad 2018, and the ITMO Open Mathematical Olympiad 2014.